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The objective of this research is to further develop the Lattice Boltzmann (LB) model in binary liquids as a computational tool to better understand convective diffusion in stochastic microgravity (g-jitter). We shall approach this problem by focusing on binary liquid alloy diffusion couples. We seek to estimate the accuracy of, or make corrections to, binary liquid diffusivities measured in microgravity. We hypothesize that the Lattice Boltzmann model might be superior to computational fluid dynamics (CFD) for computation of diffusion and fluid flow in binary liquids under the influence of stochastic microgravity. We make this conjecture because the LB model, which is based on a distribution function, is intrinsically statistical and therefore more compatible with stochastic forces. The binary liquid diffusion couple will serve as a prototype problem for other experiments that have been or will be conducted in microgravity, for example, solidification (crystal growth) from binary melts. This research is likely to result in powerful computational methods to analyze the effects of g-jitter on various space experiments that involve convective diffusion in liquids. This

Task Description:	will require careful reformulation of the LB model for binary fluids to account for species of different mass for incompressible fluids whose density is a function of composition. We will use Finite Difference Lattice Boltzmann models, rather than Lattice Gas Lattice Boltzmann models, in order to allow the space lattice to be decoupled from the velocity discretization. Chapman-Enskog expansions will be used to identify the viscosity and the diffusivity and to understand and control their dependence on composition. The model will be tested against known solutions, including solutions for non-solenoidal flow and periodic g-jitter that we have obtained previously by perturbation theory and CFD. Then it will be used to explore thoroughly the effect of periodic g-jitter on diffusivities measured in microgravity. Stochastic g-jitter will be treated to ascertain its effect on measured diffusivities. We shall also use the LB model to explore boundary conditions, such as the no-slip condition for a binary fluid, at container walls. Further development of the LB model will also provide a strong basis for the analysis of complex fluid phenomena, such as two-phase flows that are important in space engineering systems, e.g., heat exchangers. This research will be conducted by the Principal Investigator in cooperation with Dr. Victor Sofonea, an expert in computer simulation of Lattice Boltzmann models. This research is likely to result in powerful computational methods to analyze the effects of g-jitter on various space experiments that involve convective diffusion in liquids. The development of diffuse reflection boundary conditions for LB models will provide another computational tool for modeling of fluid flow in rarefied atmospheres, such as those that may be encountered by a space plane, or in microelectromechanical systems (MEMS) in the transition regime to high Knudsen numbers. This model can also be adapted to study the nonlinear dynamics of the displacement of one nearly immiscible fluid by another, such as gives rise to Safffan-Taylor instabilities in a Hele-Shaw cell.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Further development of the LB model will also provide a strong basis for the analysis of complex fluid phenomena, such as two-phase flows that are important in space engineering systems, e.g., heat exchangers. The development of diffuse reflection boundary conditions for LB models will provide another computational tool for modeling of fluid flow in the transition regime to high Knudsen numbers, such as occur in microelectromechanical systems (MEMS) . This model can also be adapted to study the nonlinear dynamics of the displacement of one nearly immiscible fluid by another, such as gives rise to Saffan-Taylor instabilities in a Hele-Shaw cell.
Task Progress:	We have adapted our Lattice Boltzmann model of binary fluids to model the displacement of one nearly immiscible fluid by another. This is a classical problem for fluids between narrowly spaced plates, known as a Hele-Shaw cell. It is well known that instabilities, known as Saffman-Taylor instabilities, occur whenever one fluid displaces a second more viscous fluid. Subsequent to instability, fingering patterns occur and the dynamics becomes nonlinear. This problem therefore poses a stringent test of our model as well as having the potential to shed light on the nonlinear dynamics. Our previous LB model of a binary solution can be modified to treat nearly immiscible fluids by including a strongly repulsive potential between A and B atoms. We actually use a regular solution model with a repulsive potential. This gives rise to a miscibility gap between an A-rich solution, alpha, and a B-rich solution, beta. We adjust parameters so that there is very little solubility of B in alpha and very little solubility of A in beta. We employ a gradient energy proportional to control the width of the thin diffuse interface between alpha and beta. The Helmholtz free energy per unit volume is expressed as a function of the number density of particles and the mole fraction of B and we derive effective forces on A and B from it. In order to treat an essentially two-dimensional flow between thin plates at spacing, we also introduce a drag force that was derived by averaging the macroscopic equations for fluid dynamics over the thin direction between plates. A and B atoms are otherwise assumed to be the same except their Boltzmann equations have different relaxation times. Since viscosity is proportional to the relaxation time, we are able to make the viscosity of alpha less than the viscosity of beta by using a relaxation time that depends linearly on composition. We tested the model for equilibrium by comparing LB simulations with results from thermodynamics for the miscibility gap. The agreement is good provided that the temperature is not too far below the critical temperature where the alpha and beta phase fields become comparable to the interface thickness between alpha and beta. We conducted LB simulations by using this model in a “linear geometry" for which a one-dimensional flow with uniform velocity is imposed at the inlet (alpha) and outlet (beta) of a rectangular domain. The simulations are preformed in a moving frame of reference. Were it not for the Saffman-Taylor instability, one would expect to observe a planar interface separating the alpha and beta phases. However, such an interface is unstable so we first observe small amplitude sinusoidal waves, as would be expected from linear stability theory. We use reflection boundary conditions on the walls of the rectangular domain that are parallel to the main flow. This imposes a relationship between the wavelength of the instability and the domain width such that only a half integral number of wavelengths can fit. Thus we cannot determine the “natural" value of the first observable wavelength very accurately. Use of large widths would suggest the use of wide domains except then we would encounter prohibitively long simulation times for the same spatial grid size. As we do simulations for increasing width, the wavelength first increases and then readjusts by half a wavelength. From data at various widths we use a least squares fit to get an approximate value of the wavelength of the first observable unstable mode, which turns out to be within about 20% of that calculated from linear stability analysis of a oversimplified model of strictly immiscible fluids that evolve quasistatically via Darcey's law. To further compare with linear stability theory, we performed a series of simulations for a domain width that varies in the vicinity of half of the estimated natural wavelength. This small domain allows us to refine the grid considerably and thus get greater accuracy. We computed the logarithmic growth rate of such Fourier modes as a function of wavelength which resulted a dispersion relation. The agreement with linear stability theory is qualitatively correct and quantitatively reasonable. The linear stability analysis is based on an oversimplified sharp-interface model in which the fluids are strictly immiscible and the dynamics are quasi-steady state, governed by Darcy's law. We are therefore reasonably confident that our model is giving reliable results. Accordingly, Alex Fore will now go on to explore its nonlinear dynamics in order to complete his doctoral dissertation, which will be published in the archival literature subsequently. Preliminary results will be presented at the APS meeting in Denver in March, 2007. Unfortunately the grant was terminated prematurely so progress did not reach its full potential.

Bibliography Type:	Description: (Last Updated: 02/21/2007)
Abstracts for Journals and Proceedings	Fore A, Sekerka R, Widom M. "A Lattice Boltzmann Model for two-dimensional flow of immiscible fluids between closely-spaced plates." Presented at the Third International Conference for Mesoscopic Methods in Engineering and Science, Hampton University, VA, July 25-28, 2006. Conference Abstracts, Third International Conference for Mesoscopic Methods in Engineering and Science, Hampton University, VA, July 2006. , Jul-2006
Articles in Peer-reviewed Journals	Sofonea V, Sekerka RF. "Boundary conditions for upwind finite difference Lattice Boltzmann model: Evidence of slip velocity in microchannel flow." Journal of Computational Physics, 2005 Aug 10;207(2):639-59. http://dx.doi.org/10.1016/j.jcp.2005.02.003 , Aug-2005
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Presentation	Sofonea, V.; Sekerka, R. F. "Lattice Boltzmann models for fluid flow at the micro/nano scale" International Workshop on Nanofluids and Magnetizable Nanocomposites, Timisoara, Romania. May-2005